Sagnik Pal

MSc in Computer Science, RKMVERI, Belur

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SUMMARY

I am a motivated and research-oriented undergraduate with a strong interest in analytical problem-solving and technical exploration. Possessing a solid foundation in machine learning and deep learning concepts, with hands-on experience building end-to-end AI and data-driven systems — from understanding core algorithms to implementing, training, and evaluating models in practical settings. I value structured learning, critical reasoning, and collaborative engagement, demonstrating a responsible and proactive approach in both academic and project environments, with a constant drive to uphold professionalism.

EDUCATION

Ramakrishna Mission Vidyamandira, Belur Math
BSc (Honours) in Computer Sceience, 2022–25
CGPA – 9.45

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)
MSc in Computer Science, 2025–27
CGPA – 7.50(till Sem 1)

SKILLS

- **Languages:** Python, JavaScript, C, C++, Java, SQL, LaTeX, php, Assembly Language
- **Frameworks:** Scikit-Learn, NumPy, Pandas, Streamlit, PyTorch, Matplotlib, Scipy, Django, Google-ADK, Google-A2A, MCP, Flask, CrewAI, NetworkX, OpenCV
- **Operating Systems:** Windows, Linux
- **Tools:** Git/GitHub, MS Office, Redis, Apache XAMPP, Linux Command Line

COURSE WORK

- Machine Learning, Artificial Intelligence, Deep Learning, Cyber Security, Emerging Technologies : LLMs, Block Chain, Mining of Massive Datasets, Data Structures
- Probability and Statistics, Linear Algebra, Computational Complexity, Discrete Mathematics, Algorithms, Compiler Design
- Operating System, Networking, Database Management System, Software Engineering
- Microprocessor and Microcontrollers, Computer Organisation and Architecture

ACHIEVEMENTS

- CUET-PG 2025 (Computer Science) – Qualified
- Secured Top 5 percentile in NPTEL course Programming In Java
- Presented "Beyond the Hub: Exploiting Structural Bridges for Targeted Epidemic Control via Graph Neural Network" at the Ajit Diwan Memorial Workshop (2026).

Certificates

- NPTEL – Programming In Java (2024)
- NPTEL – Data Base Management System (2023)
- Forage – Deloitte Australia - Technology Job Simulation (2026)

PROJECTS

Stock-LLM: Stock Market Forecasting and LLM-based Decision Support using iTrans former and Temporal Fusion Transformer (TFT)
RKMVERI Feb 2026 – May 2026
The Stock-LLM project addresses this challenge by combining deep learning-based time-series forecasting with a large language model (LLM) driven explanation layer. The project demonstrates how time-series forecasting, ensemble learning, sentiment analysis, and LLM based reasoning can be combined into one end-to-end intelligent market analytics system

NeuroGraph-X: A Hybrid CNN-GNN Framework for pMCI vs. sMCI Classification with Natural Language Explainability
RKMVERI Feb 2026 – Apr 2026
This project proposes NeuroGraph-X, a hybrid deep learning framework that integrates 3D Convolutional Neural Networks (CNNs), Graph Neural Networks (GNNs), and Natural Language Processing (NLP) for accurate and explainable AD detection from structural MRI scans. This approach aims to bridge the gap between predictive performance and medical interpretability

NetGuardAI: Containing Information Contagions A Comparative Study of Heuristic, Planning, and Deep Reinforcement Learning Strategies
RKMVERI Sep 2025 – Dec 2025
NetGuard AI is a research-oriented Reinforcement Learning framework designed to optimize the containment of misinformation (rumors) on complex social networks. By simulating SIR dynamics on modular scale-free networks, our research uncovered the "Hub Fallacy"—demonstrating that while degree-based heuristics target influencers, our GraphSAGE Agent achieves superior containment by identifying and blocking "bridge" nodes that connect isolated communities.

FaceTrack
August 2025 – October 2025
Full-stack Django application leveraging DeepFace for asynchronous video-based face authentication and multi-segment person search. The system processes uploaded videos in the background, retrieves and matches facial embeddings for identity verification of individuals across video segments.

Dynamic Client-Aware Machine Learning Model Selection in Client-Server Architectures
RKMV September 2024 – Apr 2025
This project presents a hybrid client–server framework where the server dynamically selects a machine learning model based on the client's computational capability, estimated using CPU/GPU specifications and FLOPs. The selected model is executed locally on the client, enabling efficient, scalable, and hardware-adaptive ML computation without overloading low-resource devices.

NetSCan
RKMV Feb 2024 – Mar 2024
A lightweight network scanning utility with multi-threaded support for port scanning and host status detection across local networks.